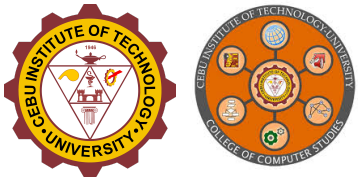
CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES



Software Project Management Plan

for

ELEMENTOPIA

Submitted by:

2627-sem1-it411-53

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1. Overview

1.1. Project Summary

1.1.1. Purpose, scope and objectives

Elementopia is an interactive, web-based gamified chemistry learning platform engineered specifically for secondary school students (primarily Grades 8 to 10 under the Philippine Department of Education K to 12 Science curriculum). Chemistry remains one of the most conceptually challenging sciences for high school learners due to abstract submicroscopic phenomena (such as valence electron configurations, ionic and covalent bonding mechanisms, and reaction stoichiometries) that cannot be directly observed in traditional lecture or rote memorization settings. Diagnostic evaluations conducted during early research indicated that 72% of secondary students experience significant difficulty in retaining and applying chemical element behaviors, while 96% expressed a clear preference for learning chemistry within a gamified interactive software environment.

This Capstone project extends an inherited baseline codebase originally developed by Rey Mar Segalle, Mark Edwin Huyo-a, Jericho Sam Dollano, Miguel Antonio Dakay, Abram John Hortezano, and Kent Bausin. The current development team (Group 53) expands and modernizes this baseline architecture by implementing three major functional modules:

1. Elemental Resonance Puzzle Module: Environmental cavern challenge chambers where students manipulate element tiles on an interactive drag-and-drop workbench to synthesize target chemical compounds, triggering chemical reactions that clear physical obstacles. It introduces a Meaningful Byproduct Diagnostic Engine (providing chemical micro-lessons instead of generic "Wrong" alerts) and an automated Hazmat Protocol Failsafe (disabling non-solution tiles after 5 failed attempts within 15 seconds to deter random guessing).

2. Assessment and Progress Module: Automated, low-latency telemetry logging (< 2 seconds) that records player compound accuracy and completion speed into a Supabase (PostgreSQL) data store. A visual, real-time Mastery Dashboard provides students with self-directed knowledge gap identification across chemical domains without requiring high-friction registration.

3. Opponent-Based Challenge Module: A competitive peer-to-peer multiplayer mode utilizing lightweight, auto-generated 5-digit room codes and temporary session nicknames to achieve rapid matchmaking (< 5 seconds). Real-time game state synchronization via Supabase Realtime (WebSockets) evaluates head-to-head Speed-to-Compound performance with 100% computational accuracy.

Project Objectives:
General Objective 1: Improve students' applied chemistry problem-solving competence to an 80% Domain success rate through interactive puzzle synthesis, reducing random guessing via the Hazmat Protocol.

General Objective 2: Automate session telemetry logging to under 2 seconds latency, enabling 90% of students to independently identify knowledge gaps via the visual Mastery Dashboard.

General Objective 3: Optimize 1v1 multiplayer matchmaking connection time to under 5 seconds, executing real-time Speed-to-Compound winner determination with 100% computational accuracy.

1.1.2. Assumptions and constraints

Assumptions:

- **Target User Infrastructure:** Secondary school students have access to consumer devices (desktop computers, laptops, Chromebooks, or mobile tablets) equipped with standard, modern web browsers (Google Chrome 110+, Mozilla Firefox 110+, Microsoft Edge 110+, or Safari 16+) supporting HTML5 Canvas and ECMAScript 6+.

- **Connectivity:** School computer laboratories and student home environments provide stable internet connectivity with minimum speeds of 1 Mbps to support WebSocket-based multiplayer state synchronization.

- **Platform Availability:** Free-tier cloud infrastructure services (Vercel for frontend hosting, Render for Spring Boot backend deployment, and Supabase for managed PostgreSQL/Realtime) provide sufficient uptime, bandwidth, and database compute quotas for educational testing cohorts of 30 to 60 concurrent users.

- **Development Continuity:** The 5-member student development team maintains uninterrupted access to development hardware, IDEs, and GitHub collaboration tools throughout the academic calendar

Constraints:

- Budgetary Constraint (Strict Freemium / Zero Budget): Total monetary project budget is constrained to ₱0.00 cash expenditure. All software libraries, build pipelines, hosting tiers, and database services must strictly leverage open-source licenses and free educational tiers.

- Schedule Constraint: The project must strictly conform to the 18-week academic semester schedule established by the College of Computer Studies, culminating in a completed deployment, Software Test Documentation (STD), and oral defense before the Capstone Examination Panel.

- Frictionless Authentication (Zero PII): To protect minor student privacy in compliance with Republic Act No. 10173 (Philippine Data Privacy Act of 2012) and maximize classroom adoption speed, the system prohibits mandatory email/password registration. Identity is established via temporary session nicknames combined with browser persisted UUID tokens.

- Curriculum Scope Bounding: Reaction mechanics are strictly bounded to high-school inorganic synthesis, binary ionic salts, simple covalent molecules, and acid-base neutralization. Advanced organic reaction mechanisms, thermodynamics, and sub-second kinetic simulations are explicitly out of scope

1.1.3. Project deliverables

All deliverables are governed by formal review and approval from the Capstone Adviser, Engr. Ralph P. Laviste, and the CCS Capstone Committee.

A. Software Deliverables

- **Fully Functional Web Application**
  - Production-ready release hosted on Vercel, integrating the refactored Resonance Puzzle, Assessment & Mastery Dashboard, and 1v1 Opponent Challenge modules.
- **Spring Boot Backend Services API**
  - Deployable JAR container hosted on Render providing RESTful reaction validation, telemetry dispatch, and failsafe monitoring.
- **Supabase Database Schema & Realtime Channels**
  - Configured PostgreSQL tables, Row-Level Security (RLS) policies, indexes, and WebSocket pub/sub channels for multiplayer duels.
- **Deployment Package and Instructions**
  - Complete source tree with automated local orchestration scripts (start-frontend.bat , start-backend.bat), Docker configurations, and environment templates (.env.example).

B. Documentation Deliverables

- **Software Requirements Specification (SRS)**
  - Comprehensive functional/non-functional specification incorporating the complete Requirements Traceability Matrix (RTM) and Post-MVP refactoring items.
- **Software Design Description (SDD)**
  - Complete architectural, component-level, database ERD, sequence, and interface specifications.
- **Software Project Management Plan (SPMP)**
  - This master managerial blueprint adhering strictly to IEEE Std 1058-1998.
- **Software Test Documents (STD Suite)**
  - Software Test Plan (STP): Scope, environment, schedule, and pass/fail thresholds.
  - Software Test Cases (STC): Detailed test specifications across unit, integration, failsafe, and multiplayer sync tests.
  - Software Test Report (STR): Empirical execution logs, pass percentages, and defect resolution summaries.
- **MVP Validation Package**
  - Empirical dataset (Elementopia (Validation Responses).xlsx), UEQ/TAM construct mapping, Teacher SME TPACK/UPA evaluation reports, and Validation Highlights Report.
- **Final Capstone Manuscript**
  - Academic thesis document formatted in accordance with CIT-U College of Computer Studies guidelines.

C. Project Presentation and Demo

- **Oral Defense Presentation Slide Deck**
  - High-impact technical presentation for the CCS Examination Panel.

- **Presentation Deck and Poster**
  - High-resolution conference poster summarizing the problem statement, system architecture, methodology, validation metrics, and conclusions.

1.1.4. Schedule and budget summary

The project follows a two-semester capstone sequence across Academic Year 2025–2026 and 2026–2027. The project timeline is structured into six major managerial phases:

Phase Milestone	Target Start Date	Target Completion Date	Primary Output / Deliverable	Status
M1: Project Initiation & Chartering	August 18, 2025	September 12, 2025	Project Charter, Proposal Defense Approval	Completed
M2: Requirements Analysis & MVP Baseline	September 15, 2025	November 28, 2025	SRS v1.0, SDD v1.0, Initial Prototype	Completed
M3: MVP Deployment & Empirical Validation	January 12, 2026	March 27, 2026	MVP Live Deployment, 65-Student Evaluation	Completed
M4: Validation Analysis & Refactoring Plan	April 06, 2026	May 29, 2026	MVP Highlights, Refactoring Priorities Form	Completed
M5: Refactoring Sprint Execution (v2.0)	August 17, 2026	October 16, 2026	Refactored Codebase, SPMP v2.0, SRS/SDD v2.0	Completed
M6: Verification, Final Testing & Defense	October 19, 2026	November 27, 2026	STD Suite, Final Oral Defense, Institutional Handover	In Progress

[Table 1.1] Schedule Allocation Plan

Budget Summary

The Elementopia project is funded under an academic freemium paradigm, utilizing institutional student licensing and open-tier infrastructure. No capital cash expenditures were disbursed.

Budget Category	Description / Specification	Commercial Cost	Actual Academic Cost	Funding Source
Frontend Hosting	Vercel Hobby Tier (Continuous Deployment, CDN, SSL)	₱20.00 / mo	₱0.00	Vercel Free Developer Tier
Backend Compute	Render Free Web Service (Spring Boot Docker instance)	₱7.00 / mo	₱0.00	Render Free Tier
Database & Realtime	Supabase Free Project (PostgreSQL, Auth, Realtime WS)	₱25.00 / mo	₱0.00	Supabase Student / Free Tier
Development IDEs	JetBrains IntelliJ IDEA Ultimate, VS Code	₱499.00 / yr	₱0.00	GitHub Student Developer Pack
Version Control & CI	GitHub Free Organization & GitHub Actions (2000 min/mo)	₱48.00 / yr	₱0.00	GitHub Educational Tier
Validation Consumables	Printing of survey instruments, evaluation tokens for SMEs	₱1,500.00	₱1,500.00	Team Self-Funded Contingency
Final Documentation	Hardbound Manuscripts & Defense Kit	₱2,500.00	₱2,200.00	Student Capstone Allocation
Total Cash Outlay	—	—	**₱3,700.00**	Student Capstone Allocation

1.2. Evolution of plan

The SPMP is reviewed at the end of each sprint and updated when scope, schedule or risks change. Changes are made through Google Docs, reviewed by the team lead, and recorded below. Updated versions are shared with the adviser.

Version	Primary Author(s)	Description of Version	Date Expected
1.0	Ocampo, Dabalos	Initial SPMP created from refactored SRS/SDD	Oct. 17, 2026
2.0	TBD	Updated after next consultation	TBD

[Table 1.2] Evolution Plan

2. References

[1] IEEE Std 1058-1998: IEEE Standard for Software Project Management Plans. IEEE Computer Society, Piscataway, NJ, 1998.

[2] IEEE Std 829-2008: IEEE Standard for Software and System Test Documentation. IEEE Computer Society, 2008.

[3] IEEE Std 1016-2009: IEEE Standard for Information Technology — Systems Design — Software Design Descriptions. IEEE Computer Society, 2009.

[4] ISO/IEC/IEEE 29148:2018: Systems and software engineering — Life cycle processes — Requirements engineering. ISO/IEC/IEEE, 2018.

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[9] Mishra, P., & Koehler, M. J. (2006): Technological pedagogical content knowledge: A framework for teacher knowledge. Teachers College Record, 108(6), 1017–1054.

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[12] Ocampo, D.M.P., Benolirao, C.C., Ceniza, J.M., Dabalos, E.Y., & Dibdib, W.K.B. (2026). Software Requirements Specification for Elementopia. CIT-U College of Computer Studies.

[13] Ocampo, D.M.P., Benolirao, C.C., Ceniza, J.M., Dabalos, E.Y., & Dibdib, W.K.B. (2026). Software Design Description for Elementopia. CIT-U College of Computer Studies.

Definitions

This clause of the SPMP shall define, or provide references to, documents containing the definition of all terms and acronyms required to properly understand the SPMP.

Term / Acronym	Full Definition & Project Context
API	**Application Programming Interface:** Structured RESTful HTTP endpoints enabling communication between the React frontend and Spring Boot backend.
Byproduct Alert	Contextual in-game micro-lesson modal triggered when an invalid element combination forms a real-world byproduct rather than the target compound.
CCB	**Change Control Board:** Governance body comprising the Project Manager, System Architect, and Capstone Adviser that evaluates scope modification requests.
CCS	**College of Computer Studies:** Academic department at Cebu Institute of Technology – University administering the Capstone program.
CIT-U	**Cebu Institute of Technology – University:** The host academic institution.
DepEd	**Department of Education (Philippines):** National executive department overseeing basic education standards, curricula, and competencies.
DoD	**Definition of Done:** Agreed-upon quality criteria required before any user story or work item can be marked as complete.
Elemental Resonance	The core puzzle mechanic wherein combining correct chemical elements synthesizes compounds that trigger physical clearance of room obstacles.
EVM	**Earned Value Management:** Project monitoring technique measuring project progress by comparing planned value, earned value, and actual cost.
Failsafe (Hazmat)	An automated scaffolding feature that disables irrelevant element tiles after detecting 5 failed synthesis attempts within 15 seconds.
GQM	**Goal-Question-Metric:** A software metrics framework linking business/educational objectives to operational telemetry and survey questions.
JWT	**JSON Web Token:** Cryptographically signed security token representing temporary session state and stateless authentication claims.
Mastery Dashboard	A personalized visual analytics panel displaying domain mastery percentages, completion speeds, and error recovery metrics.
Meaningful Byproduct	Pedagogical approach replacing binary "Wrong" alerts with accurate chemical formulas and explanatory text for accidental syntheses.
NicknameGate	Frictionless entry barrier on the frontend allowing students to enter a temporary nickname without requiring email/password registration.
PII	**Personally Identifiable Information:** Data that can identify an individual (e.g., student full name, email, phone number); explicitly omitted in Elementopia.

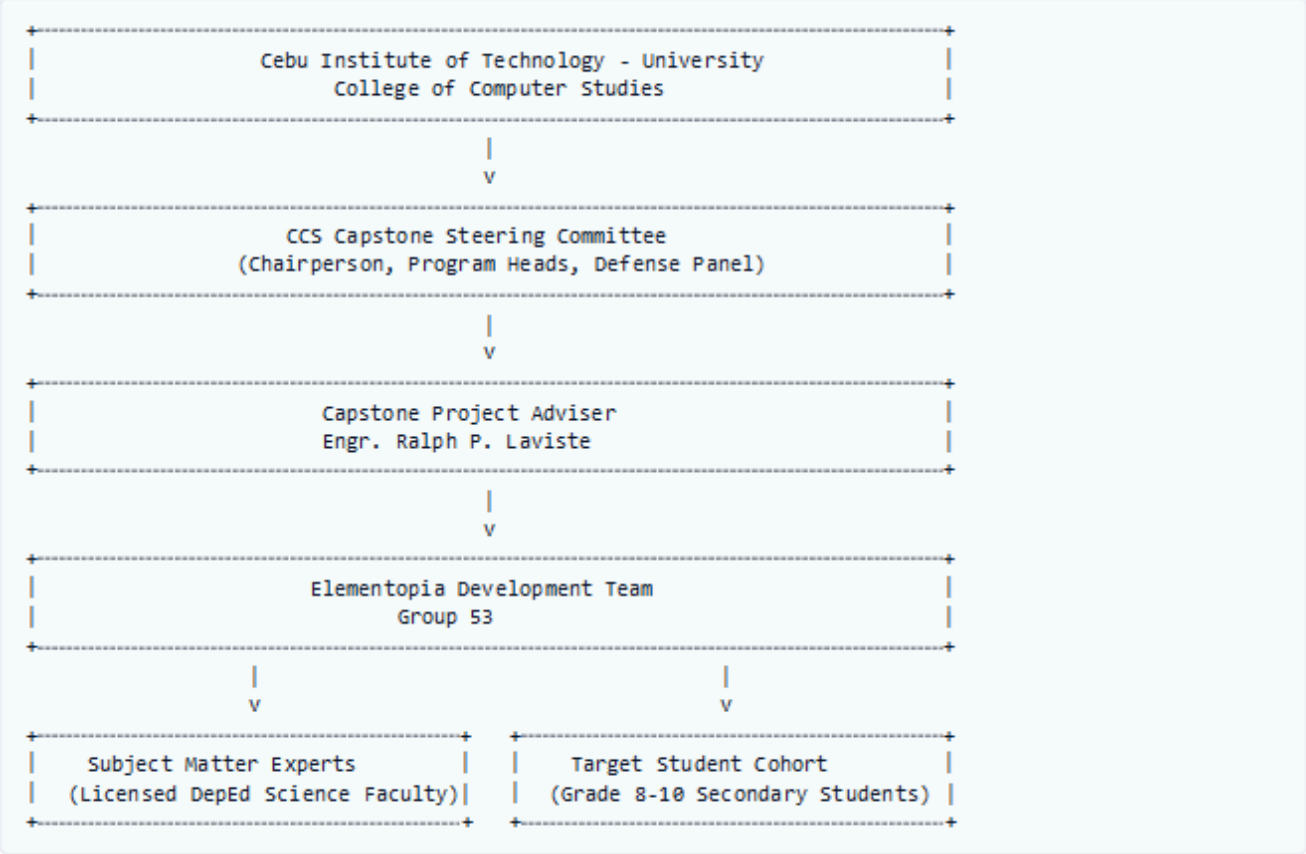
RLS	**Row-Level Security:** Database-level security policy enforced within PostgreSQL/Supabase to restrict row mutations based on session tokens.
RTM	**Requirements Traceability Matrix:** Cross-reference table ensuring bi-directional alignment between SMART objectives, user needs, system requirements, and test cases.
SDD	**Software Design Description:** IEEE 1016-compliant architecture and component design artifact.
SPMP	**Software Project Management Plan:** IEEE 1058-compliant organizational and procedural project management artifact.
SRS	**Software Requirements Specification:** ISO/IEC/IEEE 29148-compliant requirements specification artifact.
STD	**Software Test Documentation:** Formal suite comprising Software Test Plan (STP), Test Cases (STC), and Test Report (STR).
TAM	**Technology Acceptance Model:** Survey model assessing Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) to predict user adoption.

TPACK	**Technological Pedagogical Content Knowledge:** Framework evaluating teacher knowledge across technology integration, pedagogy, and chemistry content.
UEQ	**User Experience Questionnaire:** Standardized UX evaluation measuring Attractiveness, Perspicuity, Efficiency, Dependability, and Stimulation.
UPA	**Usability, Pedagogy, Accessibility:** Academic framework used by science educators to evaluate educational software viability.
UUID	**Universally Unique Identifier:** 128-bit label generated client-side and cached in `localStorage` to persist student session telemetry across visits.
WBS	**Work Breakdown Structure:** Hierarchical decomposition of the total project scope into manageable work packages and activities.

Project organization

4.1. External structure

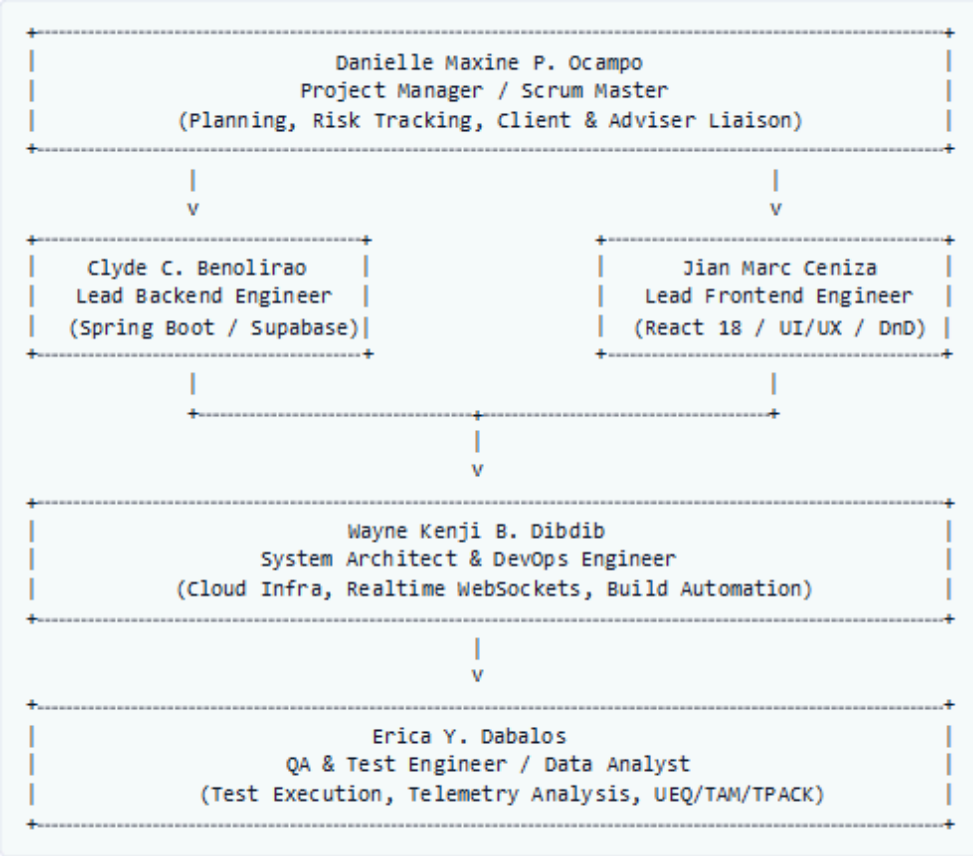
The Elementopia project functions within the governance structure of Cebu Institute of Technology – University. The external organizational hierarchy illustrates how the Capstone project team interfaces with institutional authorities, academic advisers, subject matter experts, and target student beneficiaries.



- **College of Computer Studies (CCS):** Governs course requirements, institutional policies, intellectual property guidelines, and formal milestone deadlines.
- **Capstone Steering Committee & Defense Panel:** Conducts formal evaluations at proposal defense, mid-year verification, and final oral defense.
- **Capstone Project Adviser (Engr. Ralph P. Laviste):** Provides weekly managerial oversight, technical guidance, code reviews, and milestone authorization.
- **Subject Matter Experts (SMEs):** Secondary school chemistry educators providing curriculum alignment validation and TPACK/UPA pedagogical evaluations.
- **Target Student Beneficiaries:** Junior high school chemistry students participating in empirical usability testing, skill diagnostics, and user acceptance trials.

4.2. Internal structure

The internal project organization adheres to the Agile Scrum framework tailored for an intensive 5-member undergraduate engineering unit. While team members maintain specialized functional responsibilities, crossfunctional pair-programming and collaborative testing are practiced daily.



4.3. Roles and responsibilities

To ensure accountability, each team member is assigned primary leadership and secondary support responsibilities across all project work packages.

TEAM MEMBER	ROLE & RESPONSIBILITIES	
	SEM 1	SEM 2
Ocampo	Team Lead / Project Manager	Team Lead / Project Manager
Benolirao	Frontend Developer	Frontend Developer
Ceniza	Backend Developer	Backend Developer

TEAM MEMBER	ROLE & RESPONSIBILITIES	
	SEM 1	**SEM 2**
Dabalos	Documentation Lead	Documentation Lead
Dibdib	Lead Developer	Lead Developer

[Table 4.3.1] Role and Responsibilities

Managerial process plans

5.1. Start-up plan

This subclause of the SPMP shall specify the estimation plan, staffing plan, resource acquisition plan, and training plan. Depending on the size and scope of the project, these plans may be incorporated directly or by reference to other plans.

5.1.1. Estimation plan

Estimations of development effort, sprint duration, and task complexity utilize the **Planning Poker** consensus technique combined with **Modified Wideband Delphi**.

1. **Effort Metrics (Story Points):** User stories and technical tasks in the backlog are estimated using a modified Fibonacci sequence (1, 2, 3, 5, 8, 13, 21 Story Points). One story point represents approximately 3 to 4 hours of focused student development effort.

2. **Historical Velocity Baseline:** Based on development metrics recorded during Capstone 1 (MVP phase), the team exhibits an average velocity of **32 Story Points per 2-week Sprint** across all five developers.

3. **Re-estimation Triggers:** Backlog items are formally re-estimated during Sprint Planning if:
   - A task exceeds its estimated timeline by more than 25%.
   - Technical debt in inherited code requires unforeseen architectural refactoring.
   - External adviser feedback introduces new requirements into the sprint scope.

5.1.2. Staffing plan

The project team comprises five active senior undergraduate IT students.

Team Member	Functional Specialization	Available Effort (Hours / Week)	Peak Effort Project Phase
Danielle Maxine P. Ocampo	Project Management & Documentation	18 hrs / week	Phase 1 (Initiation), Phase 4 (Validation), Phase 6 (Defense)
Clyde C. Benolirao	Spring Boot & API Engineering	20 hrs / week	Phase 2 (Architecture), Phase 5 (Refactoring Sprints)
Jian Marc Ceniza	React Frontend & UI Interaction	20 hrs / week	Phase 2 (Architecture), Phase 5 (Refactoring Sprints)
Erica Y. Dabalos	Quality Assurance & Telemetry Analysis	18 hrs / week	Phase 3 (Validation), Phase 6 (Verification & Testing)
Wayne Kenji B. Dibdib	Cloud Architecture & Realtime DevOps	20 hrs / week	Phase 3 (Deployment), Phase 5 (Realtime Optimization)

Total Available Effort	—	**96 hrs / week**	Aggregate Team Capacity

5.1.3. Resource acquisition plan

Project resources are acquired through institutional academic agreements and public open-source software distributions.

- **Development Hardware:** Team members utilize personal laptop workstations running Windows 11 and Ubuntu Linux (Intel Core i5/i7 or AMD Ryzen 5/7 processors with minimum 16 GB RAM and SSD storage).
- **Hosting & Infrastructure Services:**
- *Frontend CDN:* Vercel Hobby Tier provisioned via individual student GitHub credentials.
- *Backend Container:* Render Free Tier provisioned for Spring Boot Docker instances.
- *Database & Realtime:* Supabase Cloud instance provisioned under the free education tier.
- **Development Tools & Licenses:**
- *IDE:* IntelliJ IDEA Ultimate and WebStorm (licensed via JetBrains Student Pack).
- *Visual Studio Code:* Open-source editor equipped with ESLint, Prettier, and Spring Tools.
- *API Testing:* Postman Desktop Client (Free Collaboration Workspace).
- *Modeling: Diagrams.net and PlantUML for IEEE-compliant SDD diagrams.*
- ***Testing Hardware Acquisition:*** *Testing hardware representing student environments (4 Chromebooks, 6 Android smartphones, 4 Windows laptops) is borrowed from the CCS Capstone Laboratory for on-premise verification.*

5.1.4. Project staff training plan

To mitigate technical risk and ensure development competence across the inherited stack, targeted internal technical training sessions were conducted prior to refactoring sprints.

Training Module	Target Personnel	Delivery Method / Source	Verification / Exit Criteria
Spring Boot 3 & Java 17 Microservices	Benolirao, Dibdib	Online Guided Tutorials & Official Spring Documentation	Successful creation of a test controller with automated JUnit 5 mocks.
Supabase Realtime WebSockets	Benolirao, Ceniza, Dibdib	Supabase Documentation & WebSocket Spike Project	Creation of a prototype 2-client pub/sub message broadcaster under 200ms latency.
React 18 Drag-and-Drop & Canvas API	Ceniza, Dabalos	React DnD / HTML5 Canvas Workshops	Implementation of a functional multi-tile snap-to-grid collision workbench.
DepEd Grade 9/10 Chemistry Standards	All Team Members	Consultation with DepEd Chemistry Licensed Educators	Passing score on 20-item chemical bonding and valence assessment rubric.
Statistical Analysis for UX (UEQ & TAM)	Ocampo , Dabalos	UEQ Data Analysis Tool & TAM Regression Guidelines	Successful extraction of Cronbach's Alpha and mean benchmark scores.

5.2. Work plan

5.2.1. Work activities

5.2.1 Work Activities (Work Breakdown Structure - WBS)

The project scope is decomposed into a 4-level Work Breakdown Structure covering all managerial, technical, and supporting activities.

```
1.0 ELEMENTOPIA PROJECT MANAGEMENT & GOVERNANCE

     1.1 Project Initiation & Scope Chartering
     1.2 SPMP Formulation, Baselining & Maintenance
     1.3 Agile Sprint Cadence & Standup Governance
     1.4 Risk Monitoring & CCB Change Control
     1.5 Project Closeout & Institutional Handover

2.0 REQUIREMENTS ENGINEERING & REFACTORING SPECIFICATION
     2.1 Inherited Codebase Architectural Audit
     2.2 Post-MVP Validation Data Analysis (65 Students, 4 SMEs)
     2.3 Refactoring Priorities Specification Form
     2.4 Refactored Software Requirements Specification (SRS v2.0)
          2.4.1 DepEd Curriculum Alignment Specification
          2.4.2 Meaningful Byproduct & Failsafe Rule Definition
          2.4.3 Device UUID Session Persistence Logic
          2.4.4 Formative Skill Diagnostic Probe Requirements
2.5 Requirements Traceability Matrix (RTM) Formulation

3.0 SYSTEM DESIGN & ARCHITECTURE REFACTORING
     3.1 Refactored Software Design Description (SDD v2.0)
     3.2 Component & Class Decomposition Refactoring
     3.3 Database Relational Schema & RLS Security Redesign
     3.4 Supabase Realtime WebSocket Protocol Specification
     3.5 UI/UX Wireframing & Design System Consistency (Vanilla CSS)

4.0 CORE SOFTWARE DEVELOPMENT & MODULE IMPLEMENTATION
    4.1 Module 1: Elemental Resonance Puzzle Refactoring
        4.1.1 Interactive Drag-and-Drop Cavern Workbench (`GameBoard.jsx`)
        4.1.2 Chemical Reaction Controller & Valence Validator (`ReactionController.java`)
        4.1.3 Meaningful Byproduct Diagnostic Modals (`FeedbackLibraryService.java`)
        4.1.4 Hazmat Protocol Sliding Window Failsafe (5 fails / 15 sec)
        4.1.5 Environmental Obstacle Animations & Chemical Causality
    4.2 Module 2: Assessment & Progress Module Implementation
        4.2.1 Frictionless NicknameGate & Device UUID Session Manager (`UserService.jsx`)
        4.2.1 Supabase High-Speed Telemetry Logger (< 2 sec latency)
        4.2.3 Visual Mastery Dashboard Hub (`DashboardHub.jsx`)
        4.2.4 Post-Cavern 3-Item Formative Skill Diagnostic Probe
        4.2.5 LocalStorage Graceful Offline Degradation Handler
    4.3 Module 3: Opponent-Based Challenge Module Implementation
        4.3.1 5-Digit Room Code Generator & Lobby Management
        4.3.2 Supabase Realtime P2P Matchmaking Dispatcher (< 5 sec connection)
        4.3.3 Dual-Client Synchronized Compound Puzzle Broadcaster
        4.3.4 Speed-to-Compound Evaluation & Winner Resolver (100% Accuracy)
        4.3.5 Post-Match Results Telemetry Consolidator

5.0 VERIFICATION, TESTING & QUALITY ASSURANCE
```

```
   5.1 Software Test Plan (STP) Formulation
   5.2 Unit Testing & Mock Verification (JUnit 5 & Vitest)
   5.3 Integration Testing of Hybrid Persistence & WebSocket Sync
   5.4 End-to-End System Testing & Hazmat Failsafe Stress Testing
   5.5 User Acceptance Testing (UAT) with Target Students
   5.6 Software Test Report (STR) & Defect Remediation


6.0 SYSTEM DEPLOYMENT, ORAL DEFENSE & INSTITUTIONAL CLOSURE
   6.1 Production Cloud Provisioning (Vercel, Render, Supabase)
   6.2 User & System Administration Manual Authoring
   6.3 Final Capstone Manuscript Assembly
   6.4 Oral Defense Presentation & Live Demonstration Execution
   6.5 Codebase Archiving & Final Project Sign-Off
```

5.2.2. Schedule allocation

5.2.2 Schedule Allocation

The project schedule spans across four development sprints and two formal validation cycles.

```
Gantt Milestone Schedule (August 2026 - November 2026):
Week 01-02: [Sprint 1: Architecture Refactoring & WBS Setup]
Week 03-04: [Sprint 2: Module 1 Resonance & DepEd Chemical Alignment]
Week 05-06: [Sprint 3: Module 2 Telemetry, UUID & Formative Skill Probe]
Week 07-08: [Sprint 4: Module 3 Realtime Multiplayer & Room Matchmaking]
Week 09-10: [System Integration, Cloud Staging & Free-Tier Optimization]
Week 11-12: [Comprehensive Test Execution, UAT & STD Compilation]
Week 13-14: [Final Capstone Manuscript Finalization & Rehearsals]
Week 15   : [Final Oral Defense & Institutional Handover]
```

Critical Milestone Commitments

Milestone ID	Major Milestone Description	Target Date	Success Indicator
MS-01	Post-MVP Refactoring Priorities Form Approved	Aug 22, 2026	Formal sign-off by Engr. Ralph P. Laviste
MS-02	Refactored SRS v2.0, SDD v2.0, & SPMP v2.0 Baselined	Sep 05, 2026	Document baselined and tagged in Git
MS-03	Sprint 2 Delivery: DepEd-Aligned Resonance Workbench	Sep 19, 2026	100% pass on valence validation unit tests
MS-04	Sprint 3 Delivery: UUID Store, Telemetry & Mastery Hub	Oct 03, 2026	Telemetry write verified under 2.0s latency
MS-05	Sprint 4 Delivery: 1v1 Realtime Opponent Duel Module	Oct 17, 2026	Matchmaking connection verified under 5.0s
MS-06	Cloud Deployment Live (Vercel & Render Staging)	Oct 24, 2026	Public URL accessible with SSL

MS-07	Software Test Documentation (STD) Suite Finalized	Nov 07, 2026	Test pass rate > 95%, 0 critical defects
MS-08	Final Capstone Oral Defense & Live System Demo	Nov 20, 2026	Panel approval with passing grade

Critical Path Analysis

`2.2 Validation Analysis` -> `2.4 SRS v2.0` -> `3.1 SDD v2.0` -> `4.1.2 Reaction Controller` -> `4.2.2 Telemetry Logger` -> `4.3.2 Realtime Matchmaking` -> `5.3 Integration Testing` -> `6.1 Cloud Deployment` -> `6.4 Oral Defense` Any slippage in backend reaction logic directly delays multiplayer synchronization and subsequent testing.

5.2.3. Resource allocation

5.2.3 Resource Allocation

Resources are mapped to major WBS activities to guarantee balanced team utilization without exceeding individual 20-hour weekly thresholds.

WBS Element	Primary Lead	Secondary Support	Infrastructure / Tools Utilized
1.0 Management	Ocampo	All Members	GitHub Projects, Discord, Google Drive
2.0 Requirements	Ocampo	Benolirao, Ceniza	Google Forms, Excel Analysis Tool, DepEd Guides
3.0 Design	Dibdib	Benolirao, Ceniza	PlantUML, Diagrams.net, Postman API Designer
4.1 Module 1 (Resonance)	Ceniza	Benolirao	React 18, HTML5 Drag-Drop, Spring Boot 3
4.2 Module 2 (Assessment)	Benolirao	Ceniza, Dabalos	Supabase PostgreSQL, Chart.js, LocalStorage API
4.3 Module 3 (Opponent)	Dibdib	Benolirao, Ceniza	Supabase Realtime (WebSockets), Node.js, Render
5.0 Testing & QA	Dabalos	All Members	JUnit 5, Vitest, Postman Runner, Capstone Lab PCs
6.0 Deployment & Defense	Dibdib	Ocampo	Vercel CLI, Render Docker, LaTeX / MS Word

5.2.4. Budget allocation

5.2.4 Budget Allocation

Detailed itemization of the academic freemium operational budget:

Category	Item / Service Description	Budgeted	Actual Spent
Cloud Compute & Data	Vercel Edge Hosting (Frontend)	₱0.00	₱0.00
Cloud Compute & Data	Render Web Service (Spring Boot)	₱0.00	₱0.00
Cloud Compute & Data	Supabase PostgreSQL & Realtime WS	₱0.00	₱0.00
Domain & Networking	Vercel / Render Free Subdomains	₱0.00	₱0.00
Testing Hardware	Lab Devices (Borrowed from CIT-U)	₱0.00	₱0.00
Validation Materials	Evaluation Sheets & Tokens for SMEs	₱1,500.00	₱1,250.00
Final Documentation	Hardbound Manuscripts & Defense Kit	₱2,500.00	₱2,200.00
TOTAL ALLOCATED BUDGET		₱4,000.00	₱3,450.00

5.3. Control plan

5.3.1. Requirements control plan

5.3.1 Requirements Control Plan

Requirements creep is strictly controlled through a formal change management procedure governed by the Change Control Board (CCB), composed of Danielle Maxine P. Ocampo (Project Manager), Wayne Kenji B. Dibdib (System Architect), and Engr. Ralph P. Laviste (Adviser).

1. **Change Request Submission:** Any proposed modification to functional requirements must be submitted via a `change-request` standardized GitHub Issue labeled .
2. **Impact Analysis:** The System Architect and Lead Backend Engineer assess technical impact on the database schema, real-time latency, and sprint delivery schedule within 48 hours.
3. **CCB Evaluation:** Changes requiring more than 8 Story Points or modifying core educational objectives require unanimous CCB approval and formal sign-off by the adviser.
4. **Traceability Updating:** Upon approval, the Requirements Traceability Matrix (RTM) in SRS v2.0 is updated before code authoring begins.

5.3.2. Schedule control plan

5.3.2 Schedule Control Plan

Schedule progress is tracked dynamically through **GitHub Projects Kanban boards** and weekly velocity metrics.

- **Sprint Burndown Tracking:** Story points are tracked daily. If burndown velocity indicates a projected sprint completion lag exceeding 10%, the Scrum Master convenes an emergency triage session.

- **Descoping Strategy:** Non-critical visual enhancements (e.g., secondary particle animations or expanded compound lore cards) are designated as low priority and descoped to protect the critical path.

5.3.3. Budget control plan

5.3.3 Budget Control Plan

Because the system operates under a strict freemium constraint, budget control focuses on **Cloud Quota & Resource Usage Monitoring**:

- **Supabase DB Quotas:** Database disk usage, monthly active rows, and Realtime concurrent connections are monitored bi-weekly to prevent exceeding the 500 MB / 200 concurrent connection free-tier boundaries.
- **Render Inactivity Mitigation:** Render free-tier spin-down (hibernation after 15 minutes of inactivity) is managed by documenting a 30-second initial cold-start expectation during evaluator testing.

5.3.4. Quality control plan

5.3.4 Quality Control Plan

Quality control applies a multi-layered verification strategy across all code commits:

- **Definition of Done (DoD):** A user story is marked Done only when: (1) Code compiles with zero warnings, (2) Unit test coverage exceeds 80% for business logic, (3) Pull Request is reviewed and approved by at least one peer, (4) Automated GitHub Actions CI pipeline passes, and (5) QA verifies the feature in the staging deployment.
- **Code Reviews:** Peer reviews must adhere to the Elementopia Coding Standard (strict Airbnb JavaScript guide for React; Google Java Style Guide for Spring Boot).

5.3.5. Reporting plan

5.3.5 Reporting Plan

Communication flows are formalized across internal team units and external institutional stakeholders:

Report / Communication	Frequency	Responsible Party	Audience / Recipients	Media / Channel
Daily Asynchronous Standup	Daily (Mon–Fri)	All Team Members	Entire Team	Discord Developer Channel
Sprint Review & Demo	Bi-Weekly	Scrum Master / Devs	Team & Capstone Adviser	In-Person / Zoom Meeting
Adviser Status Briefing	Weekly	Danielle Maxine P. Ocampo	Engr. Ralph P. Laviste	Formal Status Memorandum
Institutional Progress Report	End of Month	Danielle Maxine P. Ocampo	CCS Capstone Committee	Formal University Form
Oral Defense Milestones	Mid-Year / Final	All Team Members	Capstone Panel of Examiners	Oral Defense Presentation

5.3.6. Metrics collection plan

5.3.6 Metrics Collection Plan

Project health and software quality are measured continuously through objective quantitative indicators:

Metric Category	Target Objective	Measurement Method / Tool
Sprint Velocity	>= 30 Story Points / Sprint	GitHub Projects Insights
Defect Density	< 0.5 defects per KLOC	SonarLint / GitHub Issues
Test Case Pass Rate	>= 95% test case execution success	Vitest & JUnit 5 Reports
Telemetry Latency	< 2.0 seconds end-to-end write	Supabase Log Analytics
Matchmaking Speed	< 5.0 seconds room join-to-sync	WebSocket Handshake Timers
Domain Success Rate	>= 80% puzzle solution success	Telemetry Session Data
UEQ Overall Score	>= 4.00 / 5.00 benchmark score	Standardized UEQ Analysis
Code Review Turnaround	< 24 hours from PR submission	GitHub PR Timestamps

5.3.7 Risk management plan

5.3.7 Risk Management Plan

Risk management identifies, assesses, and provides actionable mitigations for technical, operational, and institutional threats.

Risk Assessment Matrix

- **Likelihood:** Low (1), Medium (2), High (3)
- **Impact:** Low (1), Medium (2), High (3)
- **Risk Exposure Level:** Severity Score = Likelihood x Impact (High: 6–9, Moderate: 3–4, Low: 1–2)

Risk ID	Risk Category & Description	Likelihood	Impact	Severity	Mitigation & Contingency Strategy	Owner
RSK-01	**Curriculum Misalignment (DepEd):** In-game compound pool diverges from Grade 9/10 learning competencies (observed in MVP Survey Q6 = 3.77).	3	3	**9 (High)**	**Mitigation:** Execute Refactoring Priority 1; audit chemical dictionary directly against DepEd Science Curriculum Guide with 4 licensed science teachers.	Ocampo / Ceniza
RSK-02	**Free-Tier Backend Hibernation:** Render free tier spins down after 15 min of inactivity, delaying live demo by 30-50 seconds.	3	2	**6 (High)**	**Mitigation:** Implement automated keep-alive ping script 15 minutes before defense demos; display informative frontend loading state with chemistry tips.	Dibdib

RSK-03	**Nickname Collision & Session Loss:** Multiple students entering identical temporary nicknames overwrite session telemetry.	2	3	**6 (High)**	**Mitigation:** Execute Refactoring Priority 4; generate UUID stored in browser `localStorage` to namespace telemetry data regardless of nickname duplication.	Benolirao / Dibdib
RSK-04	**Multiplayer WebSocket Dropouts:** Unstable school Wi-Fi disconnects 1v1 multiplayer matches mid-challenge.	3	2	**6 (High)**	**Mitigation:** Implement Supabase Realtime reconnection heartbeat; if disconnect persists > 10s, trigger graceful forfeit screen with score preservation.	Dibdib
RSK-05	**Teacher SME Evaluator Attrition:** Licensed chemistry teachers unavailable due to DepEd quarterly examination workloads.	2	3	**6 (High)**	**Mitigation:** Secure early commitments from CIT-U High School science faculty; provide streamlined asynchronous Google Form rubrics based on TPACK/UPA.	Ocampo / Dabalos
RSK-06	**Inherited Code Technical Debt:** Undocumented legacy JavaScript routines cause unhandled exceptions during drag-and-drop.	2	2	**4 (Med)**	**Mitigation:** Perform modular refactoring of `GameBoard.jsx` and isolate game state management into dedicated React hooks; wrap in Error Boundaries.	Ceniza
RSK-07	**Academic Timeline Compression:** Simultaneous academic demands compromise team sprint velocity.	2	2	**4 (Med)**	**Mitigation:** Maintain 2-week buffer prior to final defense; cross-train developers to ensure zero single-point-of-failure across modules.	Ocampo

5.3.8 Project closeout plan

5.3.8 Project Closeout Plan

Project closeout establishes orderly decommission and institutional handover:

1. **Source Code & Asset Archiving:** Complete repository frozen and archived on the official GitHub organization with signed release tag `v2.0-final-capstone`
2. **Cloud Credentials & Environment Handover:** Cloud configuration guides, environment variables, and master administrative access documented and transferred to CCS Capstone archives.
3. **Post-Mortem Retrospective:** Team conducts a formal post-mortem debriefing documenting lessons learned, architectural insights, and recommendations for future student extension teams.
4. **Final Deliverable Submissions:** Submission of hardbound approved manuscripts, software media kits, and oral defense clearance certificates signed by Engr. Ralph P. Laviste and the examination panel.

6. Technical process plans

6.1 Process Model

The Elementopia project applies an **Agile Scrum** development methodology tailored to academic capstone realities. Development is divided into structured **2-Week Sprints**, balancing rapid feature iteration with the rigorous documentation demands of IEEE standards.



1. Sprint Planning (Day 1): Team reviews product backlog, prioritizes user stories derived from the Refactoring Master Checklist, and commits to a 2-week sprint goal.

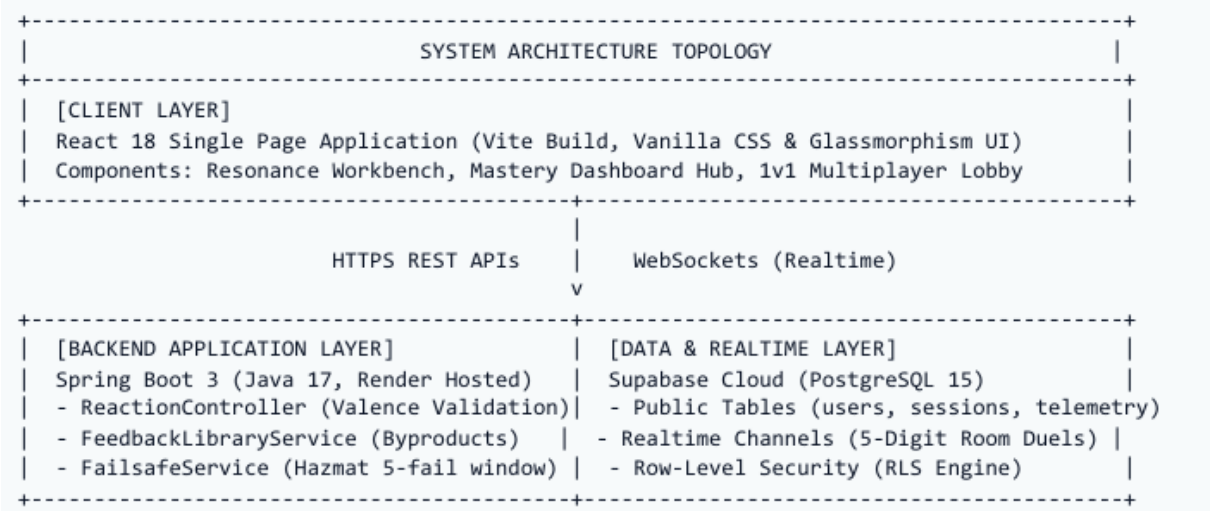
2. Daily Development & Continuous Testing (Days 2–9): Cross-functional pair programming, feature branch commits, automated unit testing, and async daily status check-ins.

3. Sprint Review & Demo (Day 10): Functional software increments are demonstrated in staging; feedback from adviser Engr. Ralph P. Laviste is incorporated into the backlog.

4. Sprint Retrospective (Day 10): Team evaluates process efficiency, velocity variances, and implements corrective actions for subsequent sprints

6.2 Methods, tools, and techniques

Development utilizes modern, enterprise-grade open-source web technologies and strict architectural separation between client presentation, business rules, and persistent telemetry.



[Table 6.2.1] System Architecture Topology

Category	Tool / Technology Selected	Justification & Applied Domain
Operating Systems	Windows 11 / Ubuntu Linux 22.04 LTS	Team workstation consistency; cross-platform validation.
Frontend Framework	**React.js 18** (Vite Toolchain)	Component-driven architecture, high-speed virtual DOM rendering for drag-and-drop puzzle tiles.
Frontend Styling	**Vanilla CSS3** & Glassmorphism	Custom, lightweight responsive UI; strict adherence to design requirements without bloated CSS framework dependencies.
Backend Engine	**Spring Boot 3.x** (Java 17 LTS)	High-performance enterprise REST API, strong type safety for stoichiometry and chemical valence verification.
Database & Realtime	**Supabase** (PostgreSQL 15 & Realtime)	Managed cloud relational database with native WebSocket engine for sub-second multiplayer matchmaking and state sync.
Version Control	**Git** & **GitHub Enterprise Education**	Distributed version control, pull request code reviews, issue tracking, and automated CI pipelines.
Automated Testing	**JUnit 5**, **Mockito**, **Vitest**	Comprehensive unit testing of backend reaction controllers and frontend component state logic.
API Validation	**Postman Desktop Client**	RESTful endpoint documentation, payload validation, and automated contract testing.

Deployment & Hosting	**Vercel** (Frontend) & **Render** (Backend)	Zero-cost continuous deployment hooked directly to GitHub repository branches with automated SSL.

[Table 6.3.1] Detailed Methods and Tools Breakdown

6.3 Infrastructure Plan

The deployment infrastructure ensures high availability, reproducible builds, and seamless transition from local development to cloud production.

- **Local Development Environment:**
  - Node.js v20 LTS, Java Development Kit (JDK 17 LTS), Apache Maven 3.9+.
  - Automated batch orchestration scripts (start-frontend.bat, start-backend.bat) allowing rapid single-click developer bootstrapping.
- **Staging / Preview Environment:**
  - Vercel automatically deploys ephemeral staging preview environments for every pull request opened against the develop branch.
- **Production Environment:**
  - Frontend: Hosted across Vercel Global Edge CDN, providing SSL/TLS 1.3 termination and automatic asset minification.
  - Backend: Containerized Spring Boot service running on Render Cloud Web Service, connected via secure environment secrets to Supabase.
  - Database: Supabase PostgreSQL instance located in the AWS Southeast Asia (Singapore) region (to minimize ping latency for Philippine learners (< 60ms)

6.3 Product Acceptance Plan

Formal product acceptance is governed by empirical verification against the Capstone Charter, signed off by Engr. Ralph P. Laviste and the CCS Capstone Examination Panel.

Objective Acceptance Criteria

1. **Functional Completion:** All 3 core modules achieve 100% functional completeness against the Refactored SRS v2.0 (Resonance Workbench, Meaningful Byproducts, Hazmat Failsafe, Mastery Dashboard, 1v1 Room Challenges).
2. **Performance Telemetry Standards:**
   - Supabase session telemetry write operations must execute in **< 2.0 seconds** across 95% of test trials.
   - P2P matchmaking room connection must establish in **< 5.0 seconds** over standard 4G/Wi-Fi connections.
   - Multiplayer Speed-to-Compound winner determination must exhibit **100% computational accuracy** with zero state desynchronization.
3. **Usability & Acceptance Thresholds:**
   - Overall User Experience Questionnaire (UEQ) benchmark mean score must exceed **4.00 out of 5.00** across tested student cohorts.
   - Subject Matter Expert (SME) TPACK/UPA evaluation must achieve an aggregate mean score >= **4.00 / 5.00** across Curriculum Alignment, Scientific Accuracy, and Instructional Scaffolding.
4. **Verification Evidence:** Formal delivery of the complete Software Test Documentation (STD) suite showing a test execution pass rate >= **95%** with zero unresolved Severity 1 or 2 defects.

7. Supporting process plans

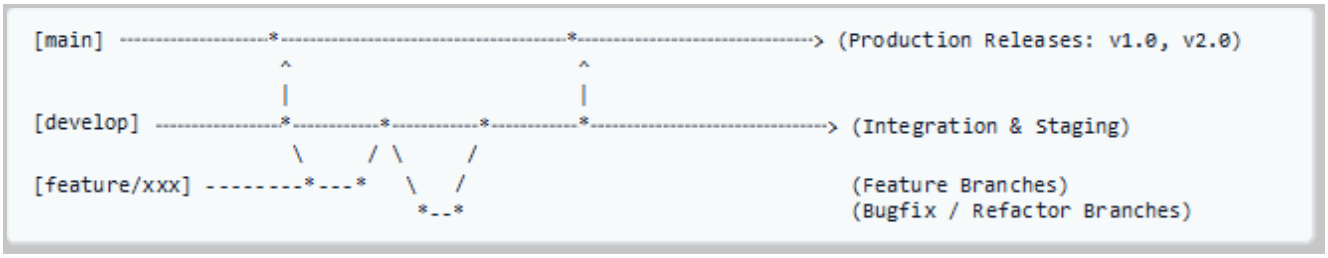
7.1. Configuration management plan

Configuration items under formal change control include:

1. Source code trees (/Elementopia/FRONTEND , /Elementopia/BACKEND).
2. Database migration scripts and Supabase schema definitions (/database/schema.sql).
3. Software engineering project documents (SPMP, SRS, SDD, STD, User Manuals).
4. Automated build scripts, Dockerfiles, and environment templates (.env.example).
5. Empirical validation datasets and evaluation survey instruments

7.1.2 Git-Flow Branching Strategy

The team strictly enforces a Git-Flow branching convention to safeguard codebase integrity:



Production-ready code only. Direct commits are strictly prohibited; requires 2-member pull request approvals

- **main :** and passing CI builds.
- **develop :** Day-to-day integration branch for sprint tasks.
- **feature/xxx :** Short-lived branches created for specific WBS user stories (e.g., feature/deped-chemical-rules , feature/uuid-persistence).
- **bugfix/xxx :** Dedicated branches for resolving defects identified during QA testing cycles.

7.2. Verification and validation plan

Verification ensures the software conforms to design specifications (building the system right), while Validation guarantees it fulfills educational learning objectives (building the right system).

Lifecycle Phase	V&V Input Artifacts	V&V Verification / Validation Tasks	V&V Output Deliverables	Responsible
Requirements Phase	Capstone Proposal, DepEd Curriculum Guides, MVP Survey Data	- Requirements traceability analysis - Curriculum alignment audit with SMEs - Ambiguity and testability reviews	- Approved SRS v2.0 • Baseline RTM • Requirements Review Report	Ocampo / Dabalos
Design Phase	Approved SRS v2.0, Inherited Architecture	- Architecture trade-off analysis - Database schema normalization audit - Component interface consistency checks	- Approved SDD v2.0 - Interface Specifications - Design Inspection Log	Dibdib / Benolirao
Implementation Phase	SDD v2.0, Source Code, Coding Standards	- Static code analysis & peer reviews - Unit test suite execution (>80% coverage) - Mocking API boundary conditions	- Unit Test Execution Logs - Clean CI Build Badges - Code Review Approvals	Benolirao / Ceniza
Integration & Test Phase	Compiled Frontend & Backend, Supabase Instances	- End-to-end integration testing - Hazmat sliding window stress testing - Realtime WebSocket latency benchmarking	- Software Test Report (STR) - Defect Remediation Log - Performance Audit	Dabalos / Dibdib
User Acceptance Phase	Deployed Staging Application, Test Scripts	- Formative skill diagnostic probe trials - Standardized UEQ/TAM survey administration - Teacher TPACK rubric evaluations	- UAT Sign-Off Certificate - Validation Highlights Report - Teacher SME Findings	Ocampo / Dabalos
Deployment Phase	Production Build Packages, Release Notes	- Post-deployment smoke testing - SSL certificate & CORS origin verification - Cold-start recovery timing	- Production Release Sign-Off - Operational Deployment Log	Dibdib

Table 7.2.2 Phase-Wise Verification and Validation (V&V) Matrix

7.3. Documentation plan

All project documentation is prepared in accordance with institutional CIT-U guidelines and IEEE software engineering documentation standards.

Document Title	Standard / Controlling Template	Primary Author	Designated Reviewer(s)	Due Date	Target Audience
Software Project Management Plan (SPMP v2.0)	IEEE Std 1058-1998	Erica Y. Dabalos	Entire Team / Engr. Ralph P. Laviste	Oct 17, 2026	Capstone Committee & Adviser
Software Requirements Specification (SRS v2.0)	ISO/IEC/IEEE 29148:2018	Erica Y. Dabalos	Clyde C. Benolirao / Jian Marc Ceniza	Oct 24, 2026	Developers, Testers, Panel
Software Design Description (SDD v2.0)	IEEE Std 1016-2009	Wayne Kenji B. Dibdib	Clyde C. Benolirao / Jian Marc Ceniza	Oct 31, 2026	Developers & Architects
Software Test Documentation Suite (STD)	IEEE Std 829-2008	Clyde C. Benolirao	Entire Team / Engr. Ralph P. Laviste	Nov 07, 2026	QA Testers, Adviser, Panel
MVP Validation Master Highlights	CIT-U CCS Capstone Guidelines	Danielle Maxine P. Ocampo	Erica Y. Dabalos	Oct 10, 2026	Science Educators & Panel
User & Administration Guide	Commercial End-User Standards	Jian Marc Ceniza	Erica Y. Dabalos	Nov 14, 2026	Students & School Instructors
Final Capstone Thesis Manuscript	CIT-U CCS Thesis Handbook	Entire Team	Engr. Ralph P. Laviste	Nov 21, 2026	Institutional Library & Panel

7.4. Quality assurance plan

Software Quality Assurance (SQA) governs adherence to technical standards, maintainability benchmarks, and pedagogical reliability throughout development.

1. **Automated Static Analysis:** All frontend code must pass ESLint and Prettier rules with zero syntax or style errors. Backend code is audited via SonarLint to identify security vulnerabilities, code smells, and cyclomatic complexity hotspots.
2. **Test Automation & Regression Gate:** Automated tests are executed via GitHub Actions on every pull request. Merges to develop are blocked if any unit or integration test fails.

3. **Traceability Auditing:** Bi-weekly audits verify that every functional requirement in the SRS traces directly to an implemented module and a corresponding test case in the STD.

7.5. Reviews and audits

Formal review gates guarantee external stakeholder visibility and institutional alignment:

- **Bi-Weekly Adviser Consultations:** Formal sprint reviews conducted with Engr. Ralph P. Laviste to demonstrate working software increments and review risk logs.
- **Code Peer Reviews:** Mandatory review of every Pull Request by a secondary developer before merging into integration branches.
- **Post-MVP Validation Audit:** Formal consultation presenting empirical findings from 65 students and securing approval for the 4 Refactoring Priorities.
- **Pre-Defense Mock Audit:** Comprehensive rehearsal of the oral defense presentation, live demonstration, and code inspection conducted one week prior to final panel examination.

7.6. Problem resolution plan

Defects, anomalous behaviors, and deviations are managed systematically through the GitHub Issues tracking infrastructure.

Severity Level	Definition / Criteria	Examples in Elementopia	SLA for Fix
Severity 1 (Blocker)	System crash, fatal reaction failure, total inability to complete puzzle rooms or establish multiplayer sync.	Synthesis engine infinite loop; WebSocket connection drop without error handling.	24 Hours
Severity 2 (Critical)	Major feature failure with no workaround; incorrect chemical stoichiometry or broken telemetry write.	Hazmat failsafe fails to disable tiles; Supabase telemetry logger drops session entries.	48 Hours
Severity 3 (Moderate)	Non-critical functional anomaly or visual defect with an available workaround.	Byproduct modal displays minor formula typo; dashboard graph rendering alignment flaw.	5 Days
Severity 4 (Minor)	Cosmetic flaw, spelling error, minor styling glitch, or minor UI hover delay.	Misaligned button margin; icon color inconsistency on high-DPI screens.	Next Sprint

Table 7.6 Defect Severity Classification & SLA

7.7. Subcontractor management plan

The Elementopia project does not employ external software subcontractors or commercial development vendors. 100% of software development, architectural refactoring, verification testing, and documentation authoring is performed by the five student developers of Group 53.

Cloud Service Vendor Governance:

Third-party cloud infrastructure providers (Vercel, Render, Supabase) are managed as software-as-a-service suppliers. The DevOps Engineer monitors vendor status portals (status.render.com , status.supabase.com) weekly to anticipate scheduled maintenance windows and ensure infrastructure uptime during testing and academic defense sessions.

7.8. Process improvement plan

Continuous process optimization is maintained throughout the project lifecycle:

- **Sprint Retrospective Insights:** At the conclusion of each 2-week sprint, the team documents *"What went well"*, *"What encountered friction"*, and *"What must change"*. Concrete improvements are incorporated immediately into the subsequent sprint plan.
- **Empirical Feedback Loop:** The 4 Refactoring Priorities established from the MVP validation prove that real user feedback directly reshapes requirements and engineering priorities, embodying true Agile process improvement.
- **Root Cause Analysis (RCA):** Any Severity 1 defect undergoes a structured 5-Whys root-cause inquiry to modify testing procedures and prevent recurrence.

8.0 Additional Plans

8.1 Student Data Privacy and Security Plan

To protect secondary school learners (minors) in strict compliance with **Republic Act No. 10173 (Philippine Data Privacy Act of 2012):**

1. Zero Personally Identifiable Information (Zero PII): The platform explicitly prohibits collecting student real names, email addresses, phone numbers, social media profiles, or geolocation data.

2. Stateless Nickname Gate & Local UUIDs: Students enter transient session nicknames. Identity is maintained client-side via a randomly generated Universally Unique Identifier (UUID) stored in browser localStorage .

3. Transport Security: All client-to-server HTTP and WebSocket communications are encrypted in transit using industry-standard TLS 1.3 / HTTPS.

4. Environment Secret Segregation: Supabase service-role keys and database passwords are encrypted within server-side environment variables and are never exposed to client-side bundles or public repositories.

8.2 Maintenance and Post-Deployment Support Plan

1. Cold-Start Keep-Alive Protocol: To manage Render free-tier hibernation, an automated health-check ping routine is activated prior to scheduled classroom testing sessions.

2. Offline-First Fallback: In the event of catastrophic school network failure, UserService.jsx gracefully stores gameplay completion flags and diagnostic probe answers in browser localStorage , preventing data loss.

3. Long-Term Academic Archiving: Following Capstone defense completion, the complete repository, automated deployment scripts, and detailed operational documentation will be archived in the CIT-U College of Computer Studies software repository for future academic cohorts.

9. Plan Annexes

WBS Code	Activity / Work Package Name	Work Description & Deliverable Specification	Assigned Resource(s)	Predecessor(s)
1.1	Project Scope Chartering	Formulate project goals, module boundaries, and secure adviser approval.	Ocampo	None
1.2	SPMP Formulation (v2.0)	Author refactored IEEE 1058 project management master plan.	Ocampo, All	1.1, 2.2
2.1	Inherited Codebase Audit	Comprehensive architectural review of legacy React & Spring Boot code.	Benolirao, Ceniza	1.1
2.2	MVP Validation Data Synthesis	Aggregate and analyze UEQ/TAM survey data from 65 students and 4 SMEs.	Dabalos, Ocampo	2.1
2.3	Refactoring Priorities Plan	Formulate official Refactoring Priorities Form across the 4 key directives.	Ocampo	2.2
2.4	Refactored SRS (v2.0)	Formulate updated requirements specification with complete RTM.	Ocampo, Ceniza	2.3
3.1	Refactored SDD (v2.0)	Formulate updated architectural diagrams, database ERDs, and sequence flows.	Dibdib, Benolirao	2.4
4.1.1	Drag-and-Drop Workbench	Refactor `GameBoard.jsx` multi-element collision and snap-to-grid canvas.	Ceniza	3.1
4.1.2	DepEd Reaction Controller	Refactor `ReactionController.java` to enforce DepEd valence and bonding rules.	Benolirao	3.1
4.1.3	Byproduct Diagnostic Engine	Implement `FeedbackLibraryService.java` providing chemistry micro-lessons.	Benolirao, Ceniza	4.1.2
4.1.4	Hazmat Failsafe Protocol	Implement sliding window algorithm (5 fails / 15s) disabling invalid tiles.	Benolirao, Ceniza	4.1.2
4.2.1	UUID Session Persistence	Implement client-side UUID persistence in `UserService.jsx` and LocalStorage.	Benolirao, Dibdib	3.1
4.2.2	Telemetry Data Dispatcher	Automate session logging to Supabase with end-to-end latency < 2.0 seconds.	Benolirao, Dibdib	4.2.1
4.2.3	Visual Mastery Dashboard	Develop `DashboardHub.jsx` displaying domain mastery analytics and badges.	Ceniza	4.2.2
4.2.4	Formative Skill Probe	Integrate 3-item post-cavern diagnostic probe assessing chemistry concepts.	Ceniza, Dabalos	4.2.3

4.3.1	5-Digit Room Code Lobby	Build matchmaking lobby generating and validating 5-digit challenge codes.	Dibdib, Ceniza	3.1
4.3.2	Realtime WebSocket Matchmaker	Implement Supabase Realtime channel broadcasting synchronized puzzles.	Dibdib, Benolirao	4.3.1
4.3.3	Speed-to-Compound Resolver	Implement 100% computationally accurate winner determination logic.	Benolirao, Dibdib	4.3.2
5.1	Software Test Plan (STP)	Author IEEE 829 test strategy, pass/fail thresholds, and test matrices.	Dabalos	2.4
5.2	Unit & Integration Testing	Execute automated JUnit 5 and Vitest test suites across all core modules.	Dabalos, All	4.1, 4.2, 4.3

Table 9.1 Annex A: Work Breakdown Structure (WBS) Dictionary

8.1 Student Data Privacy and Security Plan

Quantitative Summary of MVP Validation (Cohort: 65 Secondary Students)

- **Overall User Experience Index: 4.22 / 5.00** (High Acceptance)
- **Interactive Workbench Usability (Q4): 4.40 / 5.00** (Highest Individual Rating)
- **Multiplayer 1v1 Mode Satisfaction (Q12): 4.27 / 5.00** (Strong Engagement)
- **Curriculum Alignment Rating (Q6): 3.77 / 5.00** *(Critical Gap -> Triggered Refactoring Priority 1)*
- **Obstacle Causality Rating (Q9): 3.97 / 5.00** *(Critical Gap -> Triggered Refactoring Priority 2)*

Teacher / Subject Matter Expert (SME) Evaluation Rubric (TPACK / UPA Framework)

Administered to 4 licensed high school chemistry educators:

1. T1 (Curriculum Alignment): "Chemical elements, valence rules, and synthesized compounds align strictly with DepEd Grade 8 and 9 Science curriculum competencies (Chemical Bonding, Periodic Trends)." [Scale: 1–5]

2. T2 (Scientific Accuracy): "The formulas, reaction behaviors, and byproduct explanations are scientifically rigorous and free of conceptual fallacies." [Scale: 1–5]

3. T3 (Instructional Scaffolding): "The progression mechanic requiring foundational Element Rooms to be cleared prior to Compound Chambers provides sound pedagogical scaffolding." [Scale: 1–5]

4. T4 (Intervention Efficacy): "The Meaningful Byproduct diagnostics and Hazmat Failsafe effectively discourage blind guessing and promote guided problem-solving." [Scale: 1–5]

5. T5 (Classroom Viability): "Elementopia serves as an effective, viable complementary review tool alongside standard classroom chemistry instruction." [Scale: 1–5]

Objective ID	User Requirement (UR)	Functional Requirement (FR)	Implementation Module	Verification Test Case ID
OBJ-01	Interactive Drag-Drop Synthesis	FR-01: Canvas Workbench Collision & Snap	Module 1 (`GameBoard.jsx`)	`TC-MOD1-001`
OBJ-01	Chemistry Valence Verification	FR-02: Spring Boot Valence & Bonding Engine	Module 1 (`ReactionController.java`)	`TC-MOD1-002`
OBJ-01	Byproduct Conceptual Guidance	FR-03: Meaningful Byproduct Modal Alerts	Module 1 (`FeedbackLibraryService.java`)	`TC-MOD1-003`
OBJ-01	Anti-Guessing Intervention	FR-04: Hazmat Failsafe (5 fails in 15 sec)	Module 1 (`FailsafeService.java`)	`TC-MOD1-004`
OBJ-02	Frictionless User Identity	FR-05: NicknameGate with UUID Persistence	Module 2 (`UserService.jsx`)	`TC-MOD2-001`
OBJ-02	Low-Latency Telemetry Logging	FR-06: Telemetry Dispatch to Supabase (< 2s)	Module 2 (`TelemetryService.java`)	`TC-MOD2-002`
OBJ-02	Self-Directed Gap Analysis	FR-07: Personalized Mastery Dashboard Hub	Module 2 (`DashboardHub.jsx`)	`TC-MOD2-003`
OBJ-02	Formative Cognitive Check	FR-08: 3-Item Post-Cavern Diagnostic Probe	Module 2 (`DiagnosticProbeModal.jsx`)	`TC-MOD2-004`
OBJ-03	Rapid P2P Matchmaking	FR-09: 5-Digit Access Code Lobby (< 5s)	Module 3 (`ChallengeLobby.jsx`)	`TC-MOD3-001`
OBJ-03	Synchronized Multiplayer Duel	FR-10: Supabase Realtime State Broadcaster	Module 3 (`ChallengeController.java`)	`TC-MOD3-002`
OBJ-03	Fair Winner Determination	FR-11: Speed-to-Compound Evaluation (100%)	Module 3 (`ChallengeResolver.java`)	`TC-MOD3-003`

Table 9.3 Annex C: Requirements Traceability Matrix (RTM) Baseline Snippet

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